Same-u mixed branch composition
Independent component review

Result
The original-input same-u mixed branch composition passed fresh main compilation, producer controls, separate axiom readback and added reviewer controls. Sixty source/dependency/producer bindings were checked. No blocking discrepancy found at this field-expression scope.

Premises and source order
The theorem assumes two original normalized validTwo conditions for the source quintic, four nonzero representation weights, literal crossed support equalities, earlier inverse and equal-ordinate guards false, IsCoprime f f.derivative and2!=0. The route marker preserves support, inverse, equal/double22, zero-slope and mixed order. Earlier route markers do not claim to implement their payloads.

Composition checked
Reviewed ZeroSlope excludes sourceK=0. Cross scales transport original support and noninverse conditions to normalized coordinates. Reviewed extraction proves validPoint and nonzero z/w. Global coprimality supplies the local condition needed to derive ordinate nonzero, after which reviewed literal doubling gives nonzero output weights and validTwo. No intermediate ordinate or output-validity condition is inserted as an input assumption.

Characteristic-five witness
Producer uses f=X^5-X+1 over F5, u=X^2-1, v1=1 and v2=X with nonunit weights. Explicit Bezout certificate A=0,B=-1 holds because f derivative=-1. The two quotient witnesses X^3+X and X^3+X-1 establish original divisibility. Added reviewer checks derive extracted normalized point(1,1) and doubled normalized(q,r,s,t)=(3,1,2,4), matching u=(X-1)^2 and tangent v=2X-1.

Boundaries and failures
Equal and inverse precedence, rejected singular X^5 coprimality and characteristic-two exclusion are retained controls. Producer failed attempts remain immutable. Three additional reviewer attempts failed on opaque finite-field division reduction before explicit nonzero-denominator rewriting passed; these failures and final linter warnings are retained. Main axiom readback lists only propext, Classical.choice and Quot.sound.

Independence and scope
Reviewer authored imported extraction, doubling and ZeroSlope components; this is an independent review of the new composition, not independent authorship of those dependencies. No class-sum, UInt64 guard reflection, compiler/counter correctness or complete degree2+2 dispatch claim is inferred. No scientific trial was run; full-cohort cost ratio remains null.